

IDENTIFICATION

Identification of studies via databases and registers

Records identified via snowball retrieval
(Semantic Scholar Academic Graph API, Scopus,
ACL Anthology) on 7 of the 9 G0 seed papers:
n = 1,150

Duplicate records removed before screening:
n = 178

Identification of studies via other methods

Pre-validated corpus G0–G6, assembled by
Undermind-assisted curation of ACL, EMNLP,
NeurIPS, ICML, MLSys and related venues:
377 records exported (6 Undermind searches, 2026-04-14)
– 34 cross-group duplicates = 343; + 9 G0 seeds
n = 352

Forward-citation snowball
(Scopus n = 74; Web of Science n = 14):
n = 88

Manual Scopus / WoS cross-validation
top-up search:
n = 34

↓ records from both identification routes enter the shared screening funnel at the stages indicated below ↓

SCREENING

972 records title-screened
(snowball route only; after deduplication)

162 records pass title screening

502 records after merge with pre-validated corpus
162 snowball-screened + 352 pre-validated – 12 cross-duplicates

464 records after enrichment and topical-relevance filter

552 records in abstract-review pool
464 enriched + 88 forward-citation snowball additions

214 KEEP

173 DEFER

810 records excluded at title screening

- 791 not relevant (731 Layer-2 keyword; 60 Layer-3 triage)
- 19 UNCERTAIN — unresolved after manual triage, not carried forward

38 records removed at enrichment

- 6 off-topic
- 32 low-citation preprints moved to a rescue file (provisionally deprioritised; later confirmed excluded at abstract review)

165 records excluded at abstract review (SKIP)
individual exclusion reasons not coded at this stage (disposition reported per PRISMA 2020 Item 16b)

ELIGIBILITY

387 reports sought for retrieval
214 KEEP + 173 DEFER

224 reports assessed for eligibility
190 from the full-text queue + 34 manual cross-validation top-up (entering at this stage)

24 reports not retrieved (full text unavailable)
173 DEFER records not carried forward to full-text eligibility assessment

101 reports excluded at eligibility

- 34 manual cross-validation top-up (none retained)
- 53 background-corpus records (retained for context only)
- 14 core-corpus records failing full-text inclusion appraisal

INCLUDED

123 studies included in review
(123 records; 120 distinct papers)

70 included via
full-text-queue assessment
(review_source = fulltext)

53 included via
second abstract-review pass
(review_source = abstract-2nd-pass)

Pre-validated corpus (n = 352) joins here, from the “other methods” route.

Forward-citation snowball (n = 88) joins here, from the “other methods” route.

Manual Scopus / WoS top-up (n = 34) joins here, from the “other methods” route.